

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

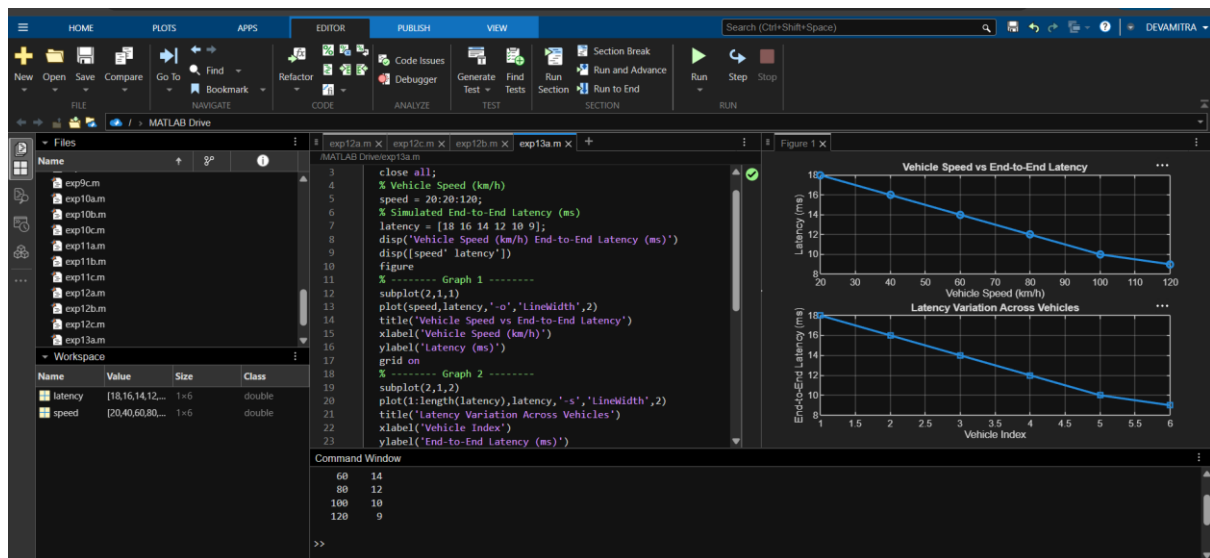
Ex.No. 13: V2X Latency Comparison Date

Objective 1 Matlab Code and Output:

Program:

```
clc;
clear;
close all;
% Vehicle Speed (km/h)
speed = 20:20:120;
% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];
disp('Vehicle Speed (km/h) End-to-End Latency (ms)')
disp([speed' latency'])
figure
% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on
% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

Output:



Objective2: Matlab Code and Output:

Program:

```
clc;

clear;

close all;

% Vehicle Speed (km/h)

speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)

PDR = [99 98 97 95 93 90];

disp('Vehicle Speed (km/h) Packet Delivery Ratio (%)')

disp([speed' PDR'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(speed,PDR,'-o','LineWidth',2)

title('Vehicle Speed vs Packet Delivery Ratio')

xlabel('Vehicle Speed (km/h)')

ylabel('Packet Delivery Ratio (%)')

grid on
```

```
% ----- Graph 2 -----
```

```
subplot(2,1,2)
```

```
plot(1:length(PDR),PDR,'-s','LineWidth',2)
```

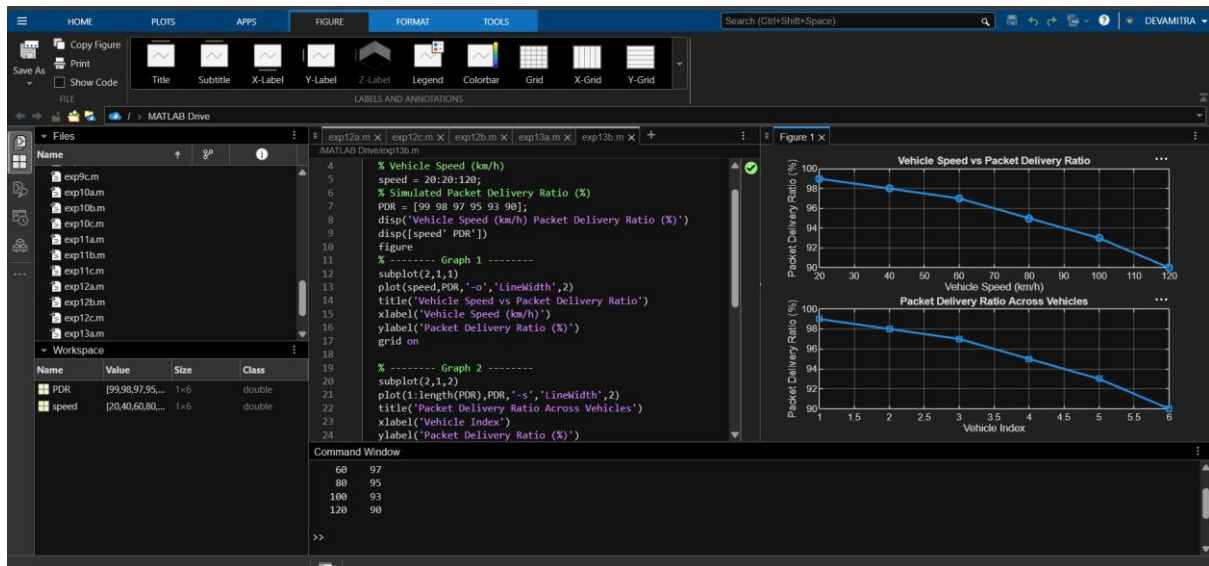
```
title('Packet Delivery Ratio Across Vehicles')
```

```
xlabel('Vehicle Index')
```

```
ylabel('Packet Delivery Ratio (%)')
```

```
grid on
```

Output:



Objective 3 Matlab Code and Output:

Program:

```
clc;
```

```
clear;
```

```
close all;
```

```
% Vehicle IDs
```

```
vehicle = 1:6;
```

```
% Vehicle Speed (km/h)
```

```
speed = [30 45 60 75 90 105];
```

```
% Communication Range (m)
```

```

range = [280 260 240 220 200 180];

disp('Vehicle Speed(km/h) Communication Range(m)')

disp([vehicle' speed' range'])

figure

% ----- Graph 1 -----

subplot(2,1,1)

plot(vehicle,speed,'-o','LineWidth',2)

title('Vehicle Speed of Different Vehicles')

xlabel('Vehicle Index')

ylabel('Vehicle Speed (km/h)')

grid on

% ----- Graph 2 -----

subplot(2,1,2)

plot(speed,range,'-s','LineWidth',2)

title('Vehicle Speed vs Communication Range')

xlabel('Vehicle Speed (km/h)')

ylabel('Communication Range (m)')

grid on

```

Output:

